

The sign-flipped pair: Theorem B and the hidden atom_{RTL} at the multiset level

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Abstract

The companion writeup `2026-05-06-twin-multiset.tex` lifted Theorem A

$$A_{(3,2,1)}(q) + A_{(5,1)}(q) = A_{(3,1,1)}(q)(A_{(3,2)}(q) + A_{(4,1)}(q))$$

to a clean multiset identity in the canonical σ_1 -G1 basis B_{2n} . Here we do the same for the sign-flipped form, Theorem B:

$$A_{(3,2,1)}(q) = A_{(3,1,1)}(q)(2A_{(3,2)}(q) - A_{(4,1)}(q)).$$

The signed combination $2M_{(3,2)} - M_{(4,1)}$ has multiplicity vector $(1, 7, 1)$ on B_4 — non-negative *and* palindromic. We name this hidden σ_1 -G1 multiset atom_{RTL} . Theorem B then becomes the multiset convolution $M_{(3,2,1)} = M_{(3,1,1)} * \text{atom}_{\text{RTL}}$, and the symmetric statement B2 introduces a non-palindromic partner $\text{atom}_{\text{LTR}} = (1, 1, 7)$ giving $M_{(5,1)} = M_{(3,1,1)} * \text{atom}_{\text{LTR}}$. Theorem C, the $2q$ -perturbation rule, drops out as the polynomial restatement: $A_{(3,2)} - A_{(4,1)} = 2q(q^2 + q + 1)$ identically. We are honest about what is missing: atom_{RTL} does not correspond to any partition of 5. Whether it is the generating function over a distinguished class of W -graph paths in $V_{(3,2,1)}$ remains open.

1 Setup

We adopt without re-derivation the framework of `2026-05-06-twin-multiset.tex` (same directory). Recall the canonical σ_1 -G1 basis

$$B_{2n} = \{ q^c(1+q)^{2n-2c} : c = 0, 1, \dots, n \} \subset \mathbb{Z}[q],$$

which is linearly independent over $\mathbb{Q}[q]$, so every polynomial in its $\mathbb{Q}$ -span has a unique expansion. For a partition $\lambda \vdash n$, the σ_1 -G1 atom $A_\lambda(q)$ expands uniquely in B_{2n} with non-negative integer coefficients,

$$A_\lambda(q) = \sum_{c=0}^n m_c^{(\lambda)} \cdot q^c(1+q)^{2n-2c},$$

and we record those coefficients as the multiplicity vector $M_\lambda = (m_0^{(\lambda)}, \dots, m_n^{(\lambda)})$, equivalently a finitely supported $\mathbb{Z}_{\geq 0}$ -valued function on $\{0, \dots, n\}$.

The convolution lemma from the companion writeup is the structural content we will lean on:

Lemma 1 (B_{2n} is closed under products; multiplication is convolution). *For all $a, b \geq 0$ and all valid c_a, c_b ,*

$$[q^{c_a}(1+q)^{2a-2c_a}] \cdot [q^{c_b}(1+q)^{2b-2c_b}] = q^{c_a+c_b}(1+q)^{2(a+b)-2(c_a+c_b)} \in B_{2(a+b)}.$$

Hence if f, g have multiplicity vectors (m_a) and (m'_b) , the product fg has multiplicity vector $(m * m')_c = \sum_{a+b=c} m_a m'_b$.

The proof is the calculation $(1+q)^{2a-2c_a} \cdot (1+q)^{2b-2c_b} = (1+q)^{2(a+b)-2(c_a+c_b)}$ together with bilinearity. See `2026-05-06-twin-multiset.tex`, Lemma 3.1.

For the rest of this paper we work with the five atoms relevant to Theorems B and C:

$$\begin{aligned} M_{(3,2,1)} &= (1, 9, 15, 2), & M_{(5,1)} &= (1, 3, 9, 14), \\ M_{(3,1,1)} &= (1, 2), & & \\ M_{(3,2)} &= (1, 5, 3), & M_{(4,1)} &= (1, 3, 5). \end{aligned} \tag{1}$$

These were extracted by the canonical greedy palindromic decomposition in `~/projects/scratch/2026-05-06-th` and re-verified May 6.

The reflected-pair phenomenon is what makes Theorem B work. Note that

$$M_{(3,2)} = (1, 5, 3) \quad \text{and} \quad M_{(4,1)} = (1, 3, 5)$$

are reflections of each other under $c \mapsto 2 - c$ on B_4 . This reflection is *not* the conjugation involution on partitions, since $(3, 2)' = (2, 2, 1) \neq (4, 1)$. It is real at S_5 , and Theorem B is its immediate consequence.

2 The hidden atom

We now isolate the structural object that powers Theorem B. The reflected pair $(1, 5, 3)$ and $(1, 3, 5)$ has the property that, in their two natural signed combinations, the cross-terms cancel into something that is palindromic on one side and concentrated on the other.

Lemma 2 (Key non-negativity).

$$2 \cdot M_{(3,2)} - M_{(4,1)} = (2, 10, 6) - (1, 3, 5) = (1, 7, 1).$$

The result is non-negative, integer-valued, and palindromic on B_4 .

Proof. Direct subtraction. The palindromicity is the equality of the first and third entries. □

Definition 3 (`atomRTL`). `atomRTL` is the σ_1 -G1 multiset on B_4 with multiplicity vector $(1, 7, 1)$. By Lemma 1 it corresponds to the polynomial

$$\text{atom}_{\text{RTL}}(q) = (1+q)^4 + 7q(1+q)^2 + q^2 = q^4 + 11q^3 + 21q^2 + 11q + 1.$$

The polynomial expansion is the calculation $(1+q)^4 = q^4 + 4q^3 + 6q^2 + 4q + 1$, $7q(1+q)^2 = 7q^3 + 14q^2 + 7q$, $q^2 = q^2$, summing to $q^4 + 11q^3 + 21q^2 + 11q + 1$.

The symmetric construction with the roles of $(3, 2)$ and $(4, 1)$ swapped produces a different multiset.

Lemma 4 (Symmetric non-negativity).

$$2 \cdot M_{(4,1)} - M_{(3,2)} = (2, 6, 10) - (1, 5, 3) = (1, 1, 7).$$

The result is non-negative but not palindromic.

Definition 5 (atom_{LTR}). atom_{LTR} is the σ_1 -G1 multiset on B_4 with multiplicity vector $(1, 1, 7)$. Its polynomial form is

$$\begin{aligned}\text{atom}_{\text{LTR}}(q) &= (1+q)^4 + q(1+q)^2 + 7q^2 \\ &= (q^4 + 4q^3 + 6q^2 + 4q + 1) + (q^3 + 2q^2 + q) + 7q^2 \\ &= q^4 + 5q^3 + 15q^2 + 5q + 1.\end{aligned}$$

The asymmetry is striking. The multiset $(1, 7, 1)$ is palindromic, the multiset $(1, 1, 7)$ is not. Both are valid σ_1 -G1 multisets in the sense that they expand non-negatively in B_4 , and both arise as integer combinations of $A_{(3,2)}$ and $A_{(4,1)}$. But only one of the two is symmetric under the natural reflection of B_4 . This asymmetry will resurface in §4.

We pause to record what these atoms are not. Neither atom_{RTL} nor atom_{LTR} is the multiset of any partition of 5: the partitions of 5 are $(5), (4, 1), (3, 2), (3, 1, 1), (2, 2, 1), (2, 1, 1, 1), (1^5)$, and one can check from the greedy expansion that none has multiplicity vector $(1, 7, 1)$ or $(1, 1, 7)$. The atoms are therefore *hidden* — they live inside the σ_1 -G1 multiset cone for B_4 but do not arise as W -graphs of irreducible S_5 -modules.

3 Theorem B at the multiset level

The signed identities of §2 promote, via the convolution lemma, to multiset identities for the two S_6 -atoms appearing in Theorem A.

Theorem 6 (B1, multiset form).

$$M_{(3,2,1)} = M_{(3,1,1)} * \text{atom}_{\text{RTL}}.$$

Theorem 7 (B2, multiset form).

$$M_{(5,1)} = M_{(3,1,1)} * \text{atom}_{\text{LTR}}.$$

Proof of Theorems 6 and 7. The polynomial Theorem B,

$$A_{(3,2,1)} = A_{(3,1,1)}(2A_{(3,2)} - A_{(4,1)}), \quad A_{(5,1)} = A_{(3,1,1)}(2A_{(4,1)} - A_{(3,2)}),$$

was verified by direct polynomial expansion (April 29, polynomial residual zero in both cases). Translating each side to its B_{2n} -multiplicity vector via Lemma 1, multiplication becomes convolution. The inner factors $2A_{(3,2)} - A_{(4,1)}$ and $2A_{(4,1)} - A_{(3,2)}$ have multiplicity vectors $(1, 7, 1) = \text{atom}_{\text{RTL}}$ and $(1, 1, 7) = \text{atom}_{\text{LTR}}$ by Lemmas 2 and 4. The conclusions are then statements about multiplicity vectors, which we tabulate directly.

For B1, with $M_{(3,1,1)} = (1, 2)$ and $\text{atom}_{\text{RTL}} = (1, 7, 1)$, the convolution is

$$\begin{aligned}(1, 2) * (1, 7, 1) &= (1 \cdot 1, \quad 1 \cdot 7 + 2 \cdot 1, \quad 1 \cdot 1 + 2 \cdot 7, \quad 2 \cdot 1) \\ &= (1, 9, 15, 2),\end{aligned}$$

which equals $M_{(3,2,1)}$.

For B2, with $\text{atom}_{\text{LTR}} = (1, 1, 7)$,

$$\begin{aligned}(1, 2) * (1, 1, 7) &= (1 \cdot 1, \quad 1 \cdot 1 + 2 \cdot 1, \quad 1 \cdot 7 + 2 \cdot 1, \quad 2 \cdot 7) \\ &= (1, 3, 9, 14),\end{aligned}$$

which equals $M_{(5,1)}$. □

It is worth pausing on the structure of these convolutions. The outer factor $M_{(3,1,1)} = (1, 2)$ is a small two-element multiset that broadcasts each entry of the inner atom, with weight 1 at offset 0 and weight 2 at offset +1. So in B1, the palindromic inner $(1, 7, 1)$ is duplicated and offset, producing the asymmetric output $(1, 9, 15, 2) = (1, 7, 1) + (0, 2, 14, 2)$. In B2, the asymmetric inner $(1, 1, 7)$ produces an even more asymmetric output $(1, 3, 9, 14)$. The asymmetry of $V_{(5,1)}$ is therefore inherited entirely from the asymmetry of atom_{LTR} — the convolution by $(1, 2)$ does not introduce new asymmetry, it only amplifies it.

4 Containment, complement, and the bijective reading

A second face of Theorem B comes into view if we refuse to absorb the signs into a single signed multiset and instead read the identity at the level of true (non-negative) multisets, with a containment and a complement.

Proposition 8 (Containment, B1 form). *$M_{(3,1,1)} * M_{(4,1)}$ is contained in $2M_{(3,1,1)} * M_{(3,2)}$ as a sub-multiset, and the complement is exactly $M_{(3,2,1)}$. Componentwise, on B_{12} :*

$$\begin{aligned} M_{(3,1,1)} * M_{(4,1)} &= (1, 5, 11, 10), \\ 2 \cdot M_{(3,1,1)} * M_{(3,2)} &= (2, 14, 26, 12), \\ (2M_{(3,1,1)} * M_{(3,2)}) - (M_{(3,1,1)} * M_{(4,1)}) &= (1, 9, 15, 2) = M_{(3,2,1)}. \end{aligned}$$

Proof. By Lemma 1 and the multiplicity vectors (1):

$$\begin{aligned} (1, 2) * (1, 3, 5) &= (1 \cdot 1, \quad 1 \cdot 3 + 2 \cdot 1, \quad 1 \cdot 5 + 2 \cdot 3, \quad 2 \cdot 5) = (1, 5, 11, 10), \\ (1, 2) * (1, 5, 3) &= (1 \cdot 1, \quad 1 \cdot 5 + 2 \cdot 1, \quad 1 \cdot 3 + 2 \cdot 5, \quad 2 \cdot 3) = (1, 7, 13, 6), \end{aligned}$$

hence $2(1, 2) * (1, 5, 3) = (2, 14, 26, 12)$, and the componentwise difference $(2, 14, 26, 12) - (1, 5, 11, 10) = (1, 9, 15, 2)$ which equals $M_{(3,2,1)}$. Non-negativity in every component is the containment. $\square$

Proposition 9 (Containment, B2 form). *$M_{(3,1,1)} * M_{(3,2)}$ is contained in $2M_{(3,1,1)} * M_{(4,1)}$ as a sub-multiset, and the complement is exactly $M_{(5,1)}$. Componentwise:*

$$\begin{aligned} M_{(3,1,1)} * M_{(3,2)} &= (1, 7, 13, 6), \\ 2 \cdot M_{(3,1,1)} * M_{(4,1)} &= (2, 10, 22, 20), \\ (2M_{(3,1,1)} * M_{(4,1)}) - (M_{(3,1,1)} * M_{(3,2)}) &= (1, 3, 9, 14) = M_{(5,1)}. \end{aligned}$$

Proof. Using the convolutions just computed: $2(1, 2) * (1, 3, 5) = 2(1, 5, 11, 10) = (2, 10, 22, 20)$, and $(2, 10, 22, 20) - (1, 7, 13, 6) = (1, 3, 9, 14) = M_{(5,1)}$. $\square$

These containments invite a bijective reading. Informally, Proposition 8 suggests an injection

$$\text{paths}(V_{(3,1,1)} \otimes V_{(4,1)}) \hookrightarrow 2 \cdot \text{paths}(V_{(3,1,1)} \otimes V_{(3,2)}),$$

preserving the c -statistic, whose image complement is in c -statistic-bijection with $\text{paths}(V_{(3,2,1)})$. Such an injection — built, presumably, from the $S_6 \downarrow S_5 \times S_1$ branching structure or from an LLT-style path operator — would upgrade Theorem 6 from a multiset identity to a W -graph path-level theorem. We do not exhibit such an injection here. It is the natural next rung.

5 Connection to Theorem C

Theorem C, the $2q$ -perturbation rule, was conjectured before the multiset structure was uncovered: it states

$$A_{(3,2)}(q) + 2q(q^2 + q + 1) = \text{atom}_{\text{RTL}}(q).$$

We can now read this as a polynomial restatement of Theorem B, via the following identity.

Lemma 10 ($2q$ -difference). $A_{(3,2)}(q) - A_{(4,1)}(q) = 2q(q^2 + q + 1)$ *identically in $\mathbb{Z}[q]$.*

Proof. Direct expansion in B_4 :

$$\begin{aligned} A_{(3,2)} - A_{(4,1)} &= [(1+q)^4 + 5q(1+q)^2 + 3q^2] - [(1+q)^4 + 3q(1+q)^2 + 5q^2] \\ &= 2q(1+q)^2 - 2q^2 \\ &= 2q[(1+q)^2 - q] \\ &= 2q(q^2 + q + 1). \end{aligned} \quad \square$$

Corollary 11 (Theorem C). $A_{(3,2)}(q) + 2q(q^2 + q + 1) = \text{atom}_{\text{RTL}}(q).$

Proof. By Lemma 10, $A_{(3,2)} + 2q(q^2 + q + 1) = A_{(3,2)} + (A_{(3,2)} - A_{(4,1)}) = 2A_{(3,2)} - A_{(4,1)}$, which equals $\text{atom}_{\text{RTL}}(q)$ by Lemma 2. $\square$

So Theorem C and Theorem B are unified. Theorem B is the multiset / structural form: it tells us that atom_{RTL} is the inner factor in the $M_{(3,1,1)}$ -convolution decomposition of $M_{(3,2,1)}$. Theorem C is the polynomial perturbation form: it tells us the same atom is reached from $A_{(3,2)}$ by adding $2q(q^2 + q + 1)$. The bridge is Lemma 10, which is the elementary observation that $A_{(3,2)}$ and $A_{(4,1)}$ differ by the symmetric polynomial $2q(q^2 + q + 1)$.

6 Discussion

The atom atom_{RTL} is hidden in two senses. It does not correspond to any partition of 5, so it does not arise as a W -graph of an irreducible S_5 -module. And it does not appear directly in any A_λ -expansion at $n \leq 5$ — it is constructed only by the signed combination $2A_{(3,2)} - A_{(4,1)}$. Yet it is a perfectly good σ_1 -G1 multiset over B_4 , palindromic, with non-negative integer entries, and it is the structural inner factor of $M_{(3,2,1)}$ at $n = 6$.

The story we now have for the $n = 5 \rightsquigarrow n = 6$ transition is the following. The $n = 5$ atoms $A_{(3,2)}$ and $A_{(4,1)}$ have multiplicity vectors $(1, 5, 3)$ and $(1, 3, 5)$, exact reflections under $c \mapsto 2 - c$ on B_4 . This reflection structure does not extend in any obvious way to $n = 6$ — the partition pair $\{(3, 2, 1), (5, 1)\}$ has no analogous reflection symmetry; their multiplicity vectors are $(1, 9, 15, 2)$ and $(1, 3, 9, 14)$, which are not reflections of each other. But the reflection at $n = 5$ leaves a fingerprint at $n = 6$. Their *sum* $(2, 8, 8)$ convolves with $(1, 2)$ into the LHS sum of Theorem A. Their *signed combinations* $(1, 7, 1)$ and $(1, 1, 7)$ — the two atoms — convolve with $(1, 2)$ into $M_{(3,2,1)}$ and $M_{(5,1)}$ individually, giving Theorem B. So Theorems A and B both descend from the same reflected pair at S_5 , with B refining A by exposing the inner structure that A averages over.

Remark 12 (An open question, preserved from earlier). Is atom_{RTL} the generating function over the closed W -graph paths of $V_{(3,2,1)}$ that traverse the s_5 -coloured edge? Or, dually, the paths that do *not* traverse it? Either reading would give a path-level interpretation of the hidden atom, and would explain why atom_{RTL} arises in $V_{(3,2,1)}$ specifically and not as the W -graph of any V_μ for $\mu \vdash 5$. We do not resolve this here.

Remark 13 (Why does atom_{RTL} generalise and atom_{LTR} does not?). The palindromicity of $\text{atom}_{\text{RTL}} = (1, 7, 1)$ is suggestive. Palindromicity on B_{2n} corresponds to the symmetry $q \leftrightarrow 1$ on the polynomial side, which is the natural duality of W -graph closed-path generating functions when one swaps ascents and descents. So atom_{RTL} behaves like a genuine W -graph-style object. $\text{atom}_{\text{LTR}} = (1, 1, 7)$ is not palindromic, and so cannot be a W -graph closed-path generating function for an unsigned graph. Pinning down the path-level interpretation of atom_{LTR} is therefore likely harder than for atom_{RTL} .

Computational verification

All multiplicity vectors (1), the signed sums of Lemmas 2 and 4, the convolutions $(1, 2) * (1, 7, 1) = (1, 9, 15, 2)$ and $(1, 2) * (1, 1, 7) = (1, 3, 9, 14)$, the containments of Propositions 8 and 9, and the polynomial difference of Lemma 10 were verified May 6, 2026 in

`~/projects/scratch/2026-05-06-theorem-B/test_theorem_B.py.`

The polynomial residuals for both B1 and B2 were 0, confirming Theorem B at the polynomial level (already known from April 29) and at the multiset level (the lift of this writeup).